# STATEMENT OF WORK

**TITLE:**
Development of the Particle Accelerator Lattice Standard (PALS) for the NP AI-Ready Accelerator Data (NARAD) effort

**TECHNOLOGY AREA:** Accelerator Technology and Control Systems

**PROJECT OBJECTIVE:**
The goal of this project is to develop the international Particle Accelerator Lattice Standard (PALS) and its extension for Machine Learning for the NP AI-Ready Accelerator Data (NARAD) effort. This will include interlaboratory consensus building on common standards, conversions of existing accelerator descriptions of BNL and JLAB machines, interfacing PALS with digital twins of these accelerators, and defining interfaces between PALS and Machine Learning applications.

**PROJECT SCOPE:**
BNL is part of the DOE-NP funded NARAD program, headed by TJNAF. BNL's scope of work includes the development and the adaptation of the PALS standard. It is this part of the scope that is being subcontracted to Cornell University by this contract.

Task 1: Interlaboratory consensus building for the PALS standard.

Task 2: Continuous extension of the PALS standard as new accelerator needs arise from the contributing laboratories, and gate keeping to maintain uniformity of the standard.

Task 3: Outreach and training to help the implementation of new standards.

Task 4: Conversions of existing accelerator descriptions of BNL and JLAB machines.

Task 5: Providing interface software for all relevant modeling tools and control system programs that need to read and write PALS files.

Task 6: Interfacing PALS with digital twins as these get developed for BNL and TJNAF accelerators.

Task 7: As NARAD intends to extend the PALS standard to the needs of Machine Learning applications, interfaces between PALS and ML/AI will be defined.

**Reports**

Regular updates will be given as presentations in the weekly NARAD meetings.

**APPLICATION TO DOE PROGRAM(S):**

This is a subcontract to the existing DOE-NP funded NARAD program.